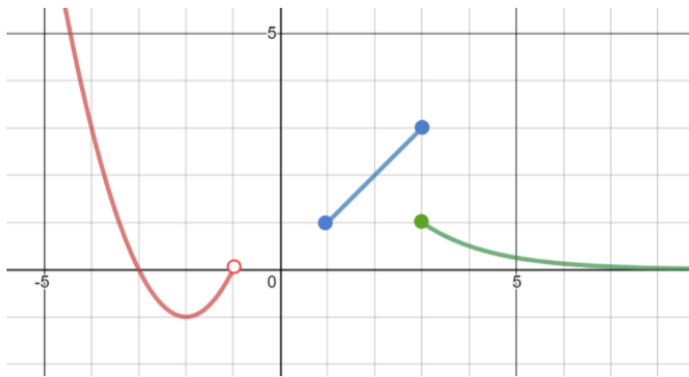


Piecewise Defined Functions – Extra Practice

1.



This is not a well-defined piece-wise defined function. Why not?

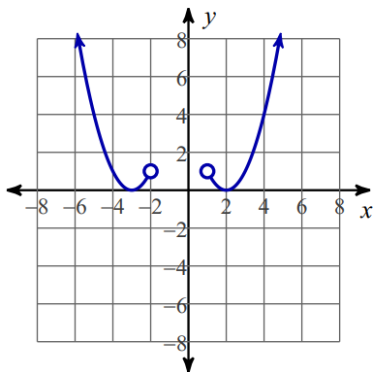
2. Consider $f(x) = \begin{cases} \frac{-1}{x}, & x < 0 \\ -\frac{1}{2}x, & 0 \leq x < 4 \\ \sqrt{x} - 2, & 4 \leq x \end{cases}$

(a). Evaluate: $f(9) =$

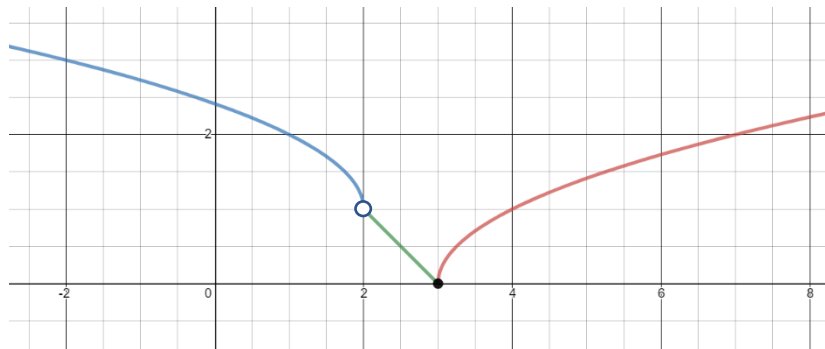
(b). Evaluate: $f(f(f(-3))) =$

3. Write a piece-wise defined function for each graph. [id what basic family; figure out what transformations to apply...]

A.



B.



4. The Geckofactor on-line store has the following shipping charges. When the weight of your total order is 5 lbs or less, there is a flat charge of \$10. If it exceeds 5 lbs but does not exceed 10lbs, each additional lb is \$1.5 /lb. If the weight is 10 lbs or more, the charge drops to \$0.80 /lb.

Write a rule for the piecewise-defined function $D(x)$ which represents the total shipping fee vs the weight x .

- What is the maximum weight of your order if you don't want to pay more than \$20 for the shipping fee? (Briefly!) Explain your reasoning.

5. Sketch your own graph of each piece-wise defined function Do not make entire tables of values (!!)

Identify the basic family. Sketch its shape. Sketch the shape with a transformation applied. THEN finalize your graph. If you want to it's ok to plug in a point right at the boundaries, but no more than that.

a. $f(x) = \begin{cases} (x-2)^2 - 1, & x < 2 \\ 2^{x-2}, & x \geq 3 \end{cases}$

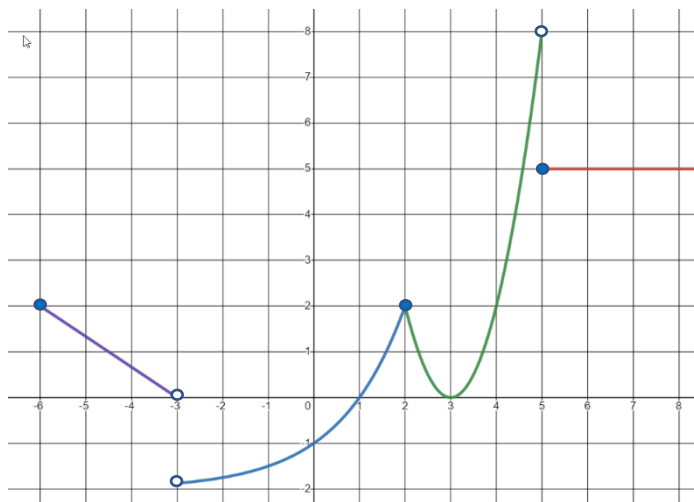
[be sure to indicate any asymptotes in your graph]

b. $g(x) = \begin{cases} 3^x + 1, & x \leq 0 \\ \frac{1}{x} + 1, & 0 < x \leq 1 \\ -x^2 - 1, & 1 < x \end{cases}$

[be sure to indicate any asymptotes in your graph]

6. This is the graph of a piece-wise defined function with 4 pieces: a linear one, an exponential one, a quadratic one, and a constant one.

(WORK SPACE)

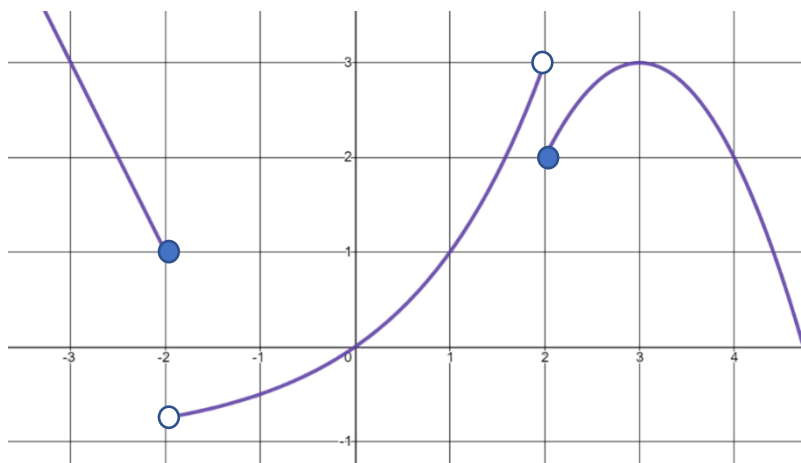


Write JUST the equation for each piece of the graph above:

- a. Linear:
- b. Exponential:
- c. Quadratic:
- d. Constant:

Combine your equation in a piecewise defined function with proper notation.

7. Write the function definition for this piece-wise function.
Be sure to use correct function notation with domain intervals.



What real value of c will make the following function continuous?

9.

$$f(x) = \begin{cases} x^2 + 2c & \text{if } x \leq 3, \\ 2cx + 1 & \text{if } x > 3. \end{cases}$$

A graph is continuous if it has no breaks. In other words, the two “pieces” of the graph have to meet.

10.

Graph the following piecewise function.

$$f(x) = \begin{cases} x^3 & \text{if } x < -1 \\ -2 & \text{if } -1 < x < 4 \\ \sqrt{x} & \text{if } x > 4 \end{cases}$$

38. $f(x) = \begin{cases} x + 1 & \text{if } x < -2 \\ -2x - 3 & \text{if } x \geq -2 \end{cases}$

39. $f(x) = \begin{cases} 2x - 1 & \text{if } x < 1 \\ 1 + x & \text{if } x \geq 1 \end{cases}$

40. $f(x) = \begin{cases} x + 1 & \text{if } x < 0 \\ x - 1 & \text{if } x > 0 \end{cases}$

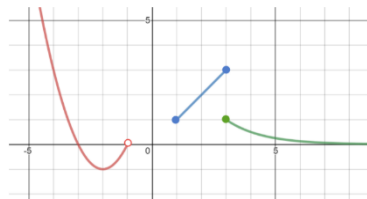
41. $f(x) = \begin{cases} 3 & \text{if } x < 0 \\ \sqrt{x} & \text{if } x \geq 0 \end{cases}$

42. $f(x) = \begin{cases} x^2 & \text{if } x < 0 \\ 1 - x & \text{if } x > 0 \end{cases}$

43. $f(x) = \begin{cases} x^2 & \text{if } x < 0 \\ x + 2 & \text{if } x \geq 0 \end{cases}$

SOLUTIONS

1.



This is not a well-defined piece-wise defined function. Why not?

There are two outputs when $x = 3$, so it can't be a function.

It's OK that the function isn't defined between -1 and 1.

2. Consider $f(x) = \begin{cases} \frac{-1}{x}, & x < 0 \\ -\frac{1}{2}x, & 0 \leq x < 4 \\ \sqrt{x} - 2, & 4 \leq x \end{cases}$

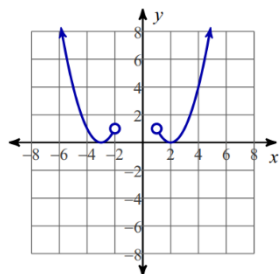
(a). Evaluate: $f(9) = 1$

(b). Evaluate: $f(f(f(-3)))$

$$f(-3) \rightarrow \frac{1}{3} \quad f\left(\frac{1}{3}\right) \rightarrow -\frac{1}{6} \quad f\left(-\frac{1}{6}\right) \rightarrow -\frac{1}{(-\frac{1}{6})} = 6$$

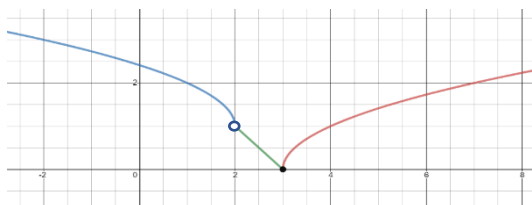
3. Write a piece-wise defined function for each graph. [id what basic family; figure out what transformations to apply...]

A



a. $f(x) = \begin{cases} (x+3)^2, & x < -2 \\ (x-2)^2, & x > 1 \end{cases}$

B.



$$g(x) = \begin{cases} \sqrt{-(x-2)} + 1, & x < 2 \\ 3 - x, & 2 < x \leq 3 \\ \sqrt{x-3}, & 3 < x \end{cases}$$

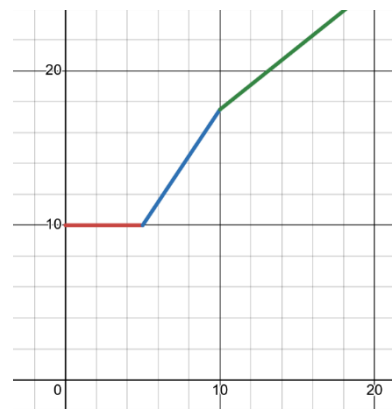
4. The Geckofactor on-line store has the following shipping charges. When the weight of your total order is 5 lbs or less, there is a flat charge of \$10. If it exceeds 5 lbs but does not exceed 10lbs, each additional lb is \$1.5 /lb. If the weight is 10 lbs or more, the charge drops to \$0.80 /lb for every additional pound beyond 10.

b. Write a rule for the piecewise-defined function $D(x)$ which represents the total shipping fee vs the weight x .

$$f(x) = \begin{cases} 10, & 0 < x \leq 5 \\ 10 + (x-5) \cdot 1.5, & 5 < x \leq 10 \\ 10 + 7.5 + (x-10) \cdot 0.8, & 10 < x \end{cases}$$

c. What is the maximum weight of your order if you don't want to pay more than \$20 for the shipping fee? (Briefly!) Explain your reasoning.

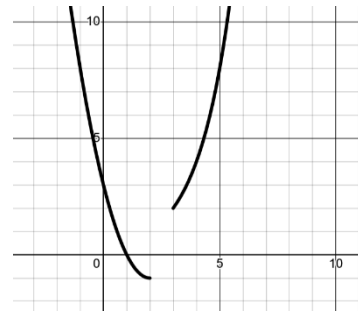
x has to be larger than 10 for the shipping to exceed \$20. We know this because $f(10) = 17.5$. So using the third equation, we need to solve: $17.5 + 0.8(x-10) = 20 \rightarrow 0.8(x-10) = 2.5 \rightarrow x-10 = 3.125 \rightarrow x = 13.125$



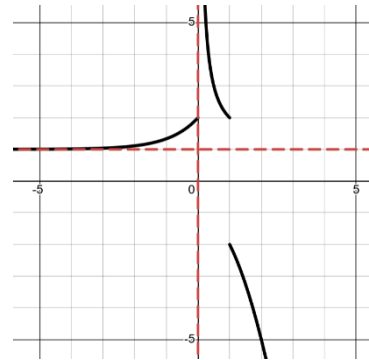
5. Sketch your own graph of each piece-wise defined function Do not make entire tables of values (!!!)

Identify the basic family. Sketch its shape. Sketch the shape with a transformation applied. THEN finalize your graph. If you want to it's ok to plug in a point right at the boundaries, but no more than that.

a. $f(x) = \begin{cases} (x-2)^2 - 1, & x < 2 \\ 2^{x-2}, & x \geq 2 \end{cases}$

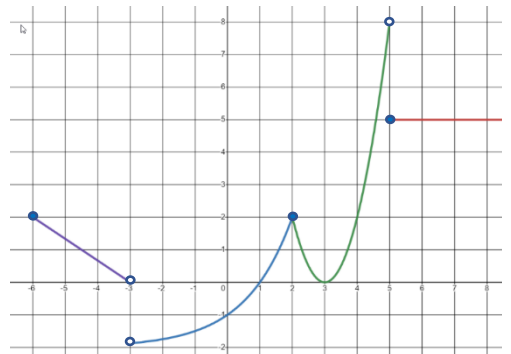


b. $g(x) = \begin{cases} 3^x + 1, & x \leq 0 \\ \frac{1}{x} + 1, & 0 < x \leq 1 \\ -x^2 - 1, & 1 < x \end{cases}$



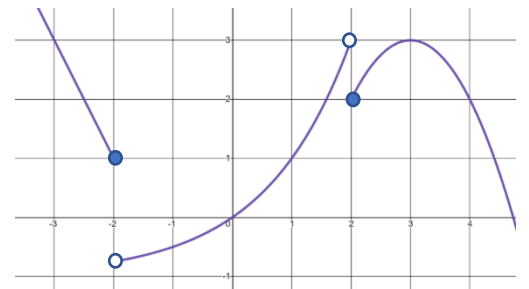
6. This is the graph of a piece-wise defined function with 4 pieces: a linear one, an exponential one, a quadratic one, and a constant one.

$$f(x) = \begin{cases} -\frac{2}{3}(x+3) & x < -3 \\ 2^x - 2 & -3 < x \leq 2 \\ 2 \cdot (x-3)^2 & 2 < x < 5 \\ 5 & x \geq 5 \end{cases}$$



7. Write the function definition for

$$g(x) = \begin{cases} -2(x+2) + 1, & x < -2 \\ 2^x - 1, & 0 < x < 2 \\ -(x-3)^2 + 3, & 2 \leq x \end{cases}$$



9.

What real value of c will make the following function continuous?

$$f(x) = \begin{cases} x^2 + 2c & \text{if } x \leq 3, \\ 2cx + 1 & \text{if } x > 3. \end{cases}$$

A graph is continuous if it has no breaks. In other words, the two "pieces" of the graph have to meet.

If the two pieces have to meet, that means both equations need to have the same y -value when $x = 3$ (because that's where they join. Let's set up an equation: $(3)^2 + 2c = 2c(3) + 1 \rightarrow 2c + 9 = 6c + 1 \rightarrow 8 = 4c \rightarrow c = 2$

These ones you can check on desmos =)

Graph the following piecewise function.

$$f(x) = \begin{cases} x^3 & \text{if } x < -1 \\ -2 & \text{if } -1 < x < 4 \\ \sqrt{x} & \text{if } x > 4 \end{cases}$$

38. $f(x) = \begin{cases} x+1 & \text{if } x < -2 \\ -2x-3 & \text{if } x \geq -2 \end{cases}$

39. $f(x) = \begin{cases} 2x-1 & \text{if } x < 1 \\ 1+x & \text{if } x \geq 1 \end{cases}$

40. $f(x) = \begin{cases} x+1 & \text{if } x < 0 \\ x-1 & \text{if } x > 0 \end{cases}$

41. $f(x) = \begin{cases} 3 & \text{if } x < 0 \\ \sqrt{x} & \text{if } x \geq 0 \end{cases}$

42. $f(x) = \begin{cases} x^2 & \text{if } x < 0 \\ 1-x & \text{if } x > 0 \end{cases}$

43. $f(x) = \begin{cases} x^2 & \text{if } x < 0 \\ x+2 & \text{if } x \geq 0 \end{cases}$